

5-gram Language Model – Jane Austen Style

Python Implementation

```
from collections import defaultdict
import random

def build_ngrams(text, n=5):
    words = text.lower().split()
    model = defaultdict(list)
    for i in range(len(words)-n):
        key = tuple(words[i:i+n-1])
        model[key].append(words[i+n-1])
    return model

def generate_text(model, seed, length=30):
    output = seed.lower().split()
    for _ in range(length):
        key = tuple(output[-4:])
        if key in model:
            output.append(random.choice(model[key]))
        else:
            break
    return " ".join(output)
```

Sample Inputs and Outputs

Input: It is a truth universally

Output: it is a truth universally acknowledged, that a single man in possession of a good fortune, must be in want of a wife. however little known the feelings or views of such a man may

Input: However little known the feelings

Output: however little known the feelings or views of such a man may be on his first entering a neighbourhood, this truth is so well fixed in the minds of the surrounding families, that he is

Input: This truth is so well

Output: this truth is so well fixed in the minds of the surrounding families, that he is considered the rightful property of someone or other of their